

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Semester Examination – Summer 2023

Course: B. Tech. Branch : E&TC and allied Semester :IV

Subject Code & Name: BTBS 404/ Probability Theory and Random Processes

Max Marks: 60 Date:26/07/2023 Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.

2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.

3. Use of non-programmable scientific calculators is allowed.

4. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks														
Q. 1	Solve Any Two of the following.		12														
A)	For a certain binary communication channel probability that transmitted '0' received as '0' is 0.95 and probability that transmitted '1' received as '1' is 0.9. if probability that '0' transmitted is 0.4. Find the probability that 1) A '1' is received 2) A '1' is transmitted given that '1' was received.	L2/CO1	6														
B)	State and explain Theorem of Total Probability. Also state Bayes' Theorem.	L1/CO1	6														
C)	From 6 positive and 8 negative numbers. 4 numbers are chosen at random (without replacement). What is the probability that product is positive.	L2/CO1	6														
Q.2	Solve Any Two of the following.		12														
A)	What are PDF and CDF. State and explain properties of CDF	L1/CO2	6														
B)	Suppose the density function of X& Y is given by $F(x,y)= 6xy(2-x-y) \ 0<x<1, \ 0<y<1$ = 0 otherwise Compute condition expectation of X given that Y=y where $0<y<1$.	L2/CO2	6														
C)	Calculate the correlation coefficient between X& Y from following data <table><tr><td>X</td><td>10</td><td>14</td><td>18</td><td>22</td><td>26</td><td>30</td></tr><tr><td>Y</td><td>18</td><td>12</td><td>24</td><td>6</td><td>30</td><td>36</td></tr></table>	X	10	14	18	22	26	30	Y	18	12	24	6	30	36	L2/CO2	6
X	10	14	18	22	26	30											
Y	18	12	24	6	30	36											
Q. 3	Solve Any Two of the following.		12														
A)	State and prove Tchebycheff Inequality.	L1/CO3	6														
B)	Bus arrived at a specific stop at 15 min interval starting from 7AM. That is they arrive at 7, 7.15, 7.30, 7.45 and so on. If passenger arrives at a stop at random time that is uniformly distributed between 7 and 7.30 AM Find the probability that he waits a) Less than 5 mins for a bus	L2/CO3	6														

	b) At least 12 mins for the bus		
C)	State and Explain properties of Covariance matrix	L1/CO3	6
Q.4	Solve Any Two of the following.		12
A)	Explain in detail with proof central limit theorem	L1/CO4	6
B)	Distinguish between WLLN and SLLN in detail.	L1/CO4	6
C)	How to determine convergence in Probability. Elaborate in brief	L1/CO4	6
Q.5	Solve Any Two of the following.		12
A)	A random process is given by $X(t) = A \cos w_0 t + B \sin w_0 t$ Where w_0 is constant. A & B are independent random variables having values -1 and 2 with probabilities 2/3 and 1/3 respectively. Find mean and auto-correlation function.	L2/CO5	6
B)	Explain in brief Strict Sense Stationary process (SSS) and Wide Sense Stationary (SSS) Process	L1/CO5	6
C)	Explain power spectral density function with its properties.	L1/CO5	6
	*** End ***		